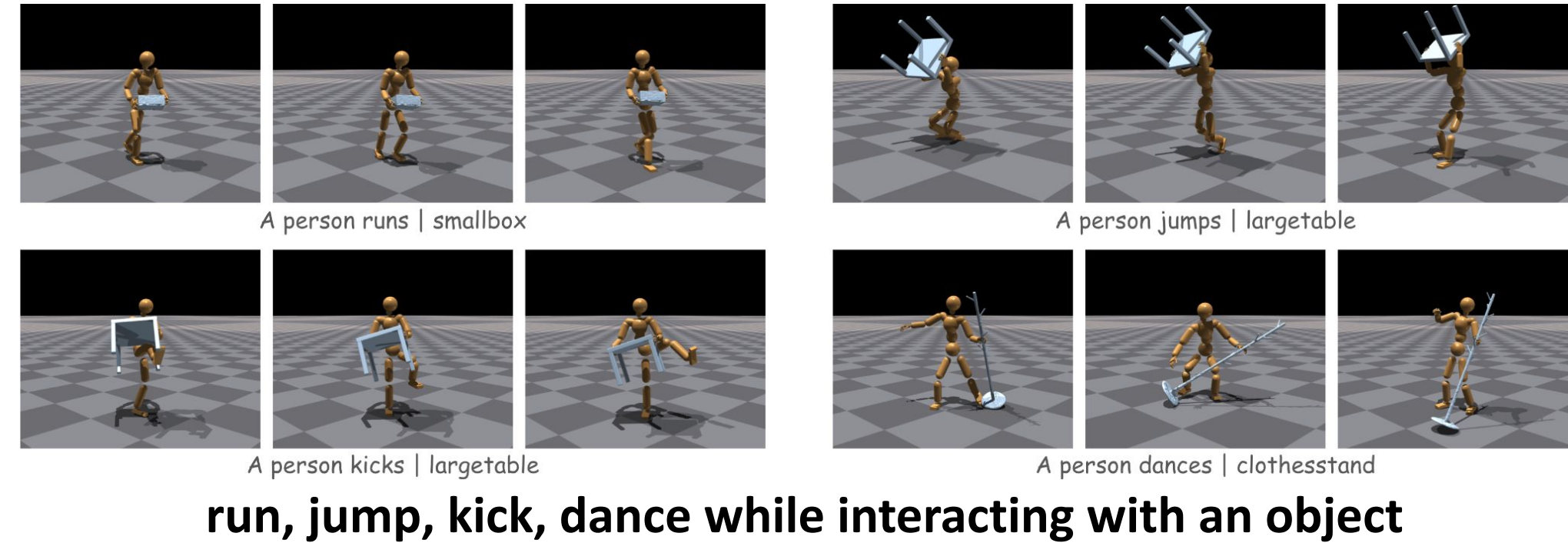


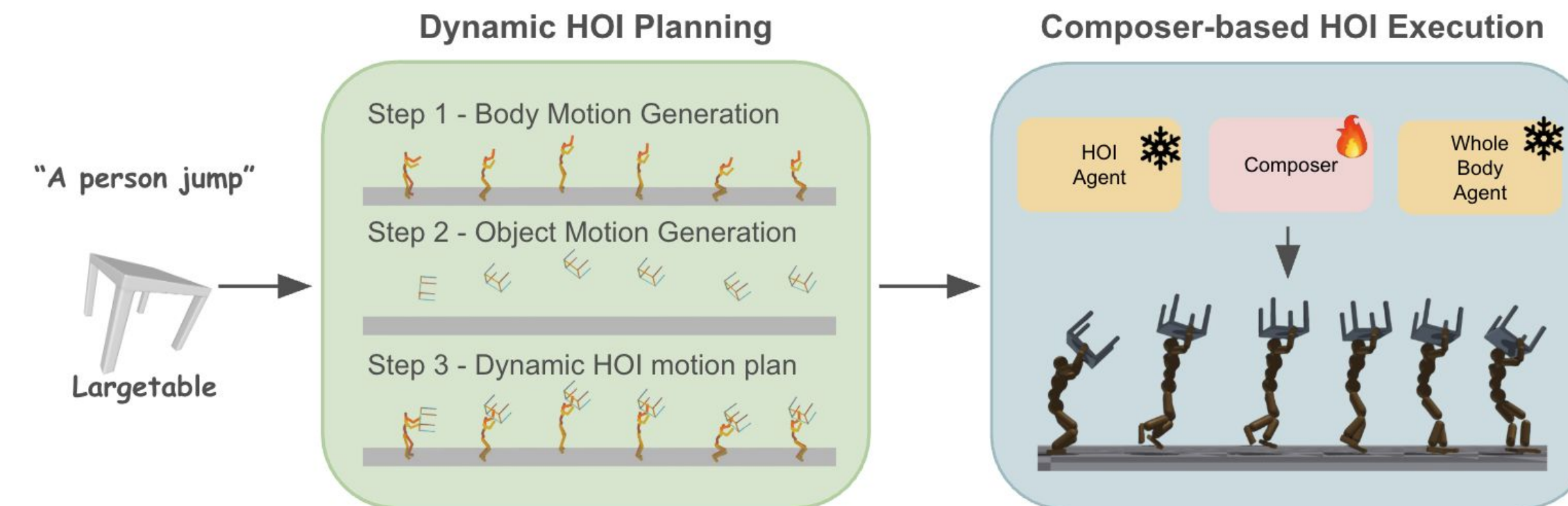


Introduction



- Plan and execute physically plausible, **dynamic** and **contact-rich Human-Object Interaction (HOI)**
- Existing datasets and agents handle either HOI or dynamic motions, but **not both simultaneously**

Overview



- Dynamic HOI Planning:** Plan dynamic motions with consistent hand-object contact
- Composer-based HOI Execution:** Efficiently learn dynamic HOI control by blending pre-trained experts

Quantitative Results

Method	HOI	Physics	HOI QUALITY		PHYSICAL PLAUSIBILITY				MOTION QUALITY	
			$C_{\%} \uparrow$	$C_{\text{cons}} \uparrow$	Pene _{obj} ↓	Skate ↓	Float ↓	Jitter _{pos} ↓	R-Prec ↑	Diversity ↑
MDM			—	—	8.18	0.133	29.3	9.43×10^4	0.374	7.61
HOI-Diff	✓		0.285	18.2	0.633	17.8	1.56×10^4	0.257	4.66	
DAViD	✓		0.848	10.9	4.842	0.261	20.4	6.69×10^4	0.310	6.70
Ours _p	✓		1.000	0.906	4.196	0.217	30.1	5.93×10^4	0.332	5.56
Ours _{p+E}	✓	✓	0.999	2.95	0.009	0.786	9.95	4.53×10^4	0.316	3.97

Generation Quality
(Ours_p: planning only / Ours_{p+E}: full planning-execution pipeline)

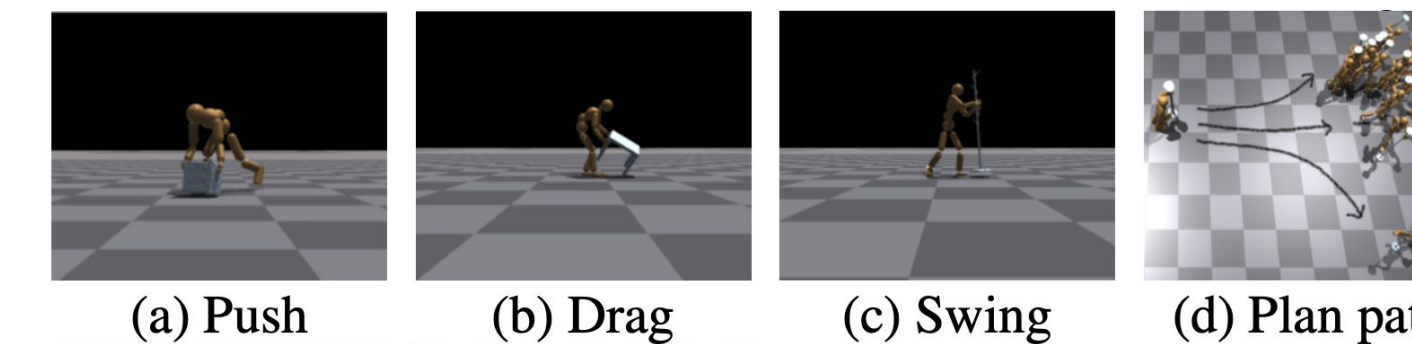
- Ours_p achieves the **best HOI Quality** — $C_{\%}$ and C_{cons}
- Ours_{p+E} achieves the **best physical plausibility** — Pene_{obj} and Float, 2nd Jitter_{pose}

Method	$SR \uparrow$	T	$D \uparrow$	$E_{\text{HOI}} \downarrow$	Jitter _{DoF} $\downarrow$				
PPO	0.227	75	0.859	59.098	0.424×10^3	Blending	SR $\uparrow$	D $\uparrow$	$E_{\text{HOI}} \downarrow$
PHC [26]	0.397	-	1.166	12.965	1.350×10^3	Heuristic _{C_{hand}}	0.365	1.069	13.0
PHC _R	0.313	15	1.834	26.024	1.762×10^3	Heuristic _{C_{arm}}	0.472	1.603	17.1
InterMimic [56]	0.376	-	2.407	21.379	0.436×10^3	Hard MoE	0.383	2.748	20.4
InterMimic _R	0.310	15	2.324	34.675	0.336×10^3	Hard MoE (Joint)	0.383	2.535	26.4
InterMimic _{FT}	0.526	75	7.37	8.635	0.370×10^3	Ours _{MLP}	0.571	3.034	18.0
Ours	0.591	23	<u>4.607</u>	<u>11.667</u>	0.359×10^3	Ours _{MLP+PCA}	0.591	4.607	11.6

Execution Performance
(Ours_{MLP}: except PCA subspace exploration)

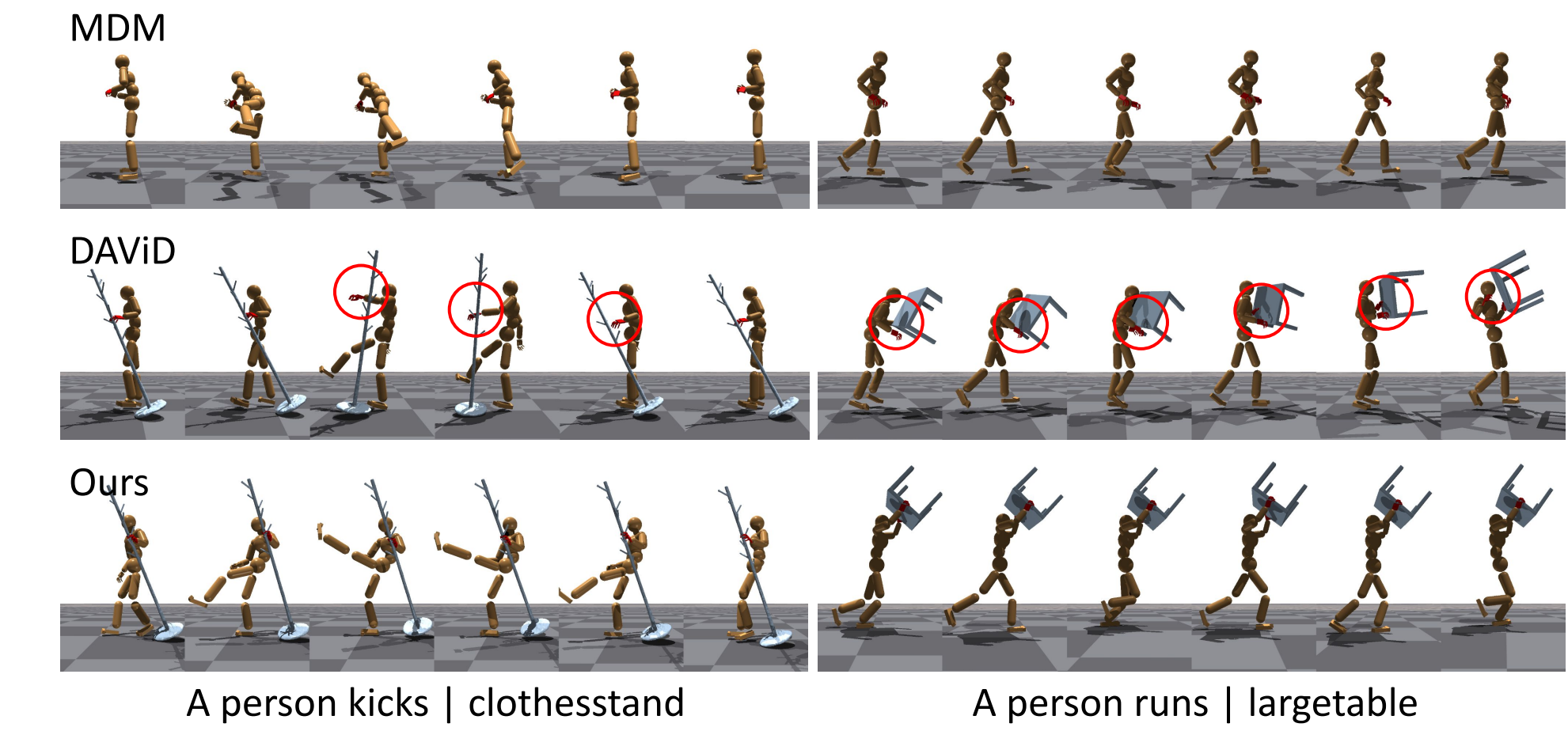
- Ours achieves the **highest success rate**
- Converges **3× faster** than fine-tuning InterMimic
- Ours blending method achieved best SR, D, and E_{HOI}

Scalability



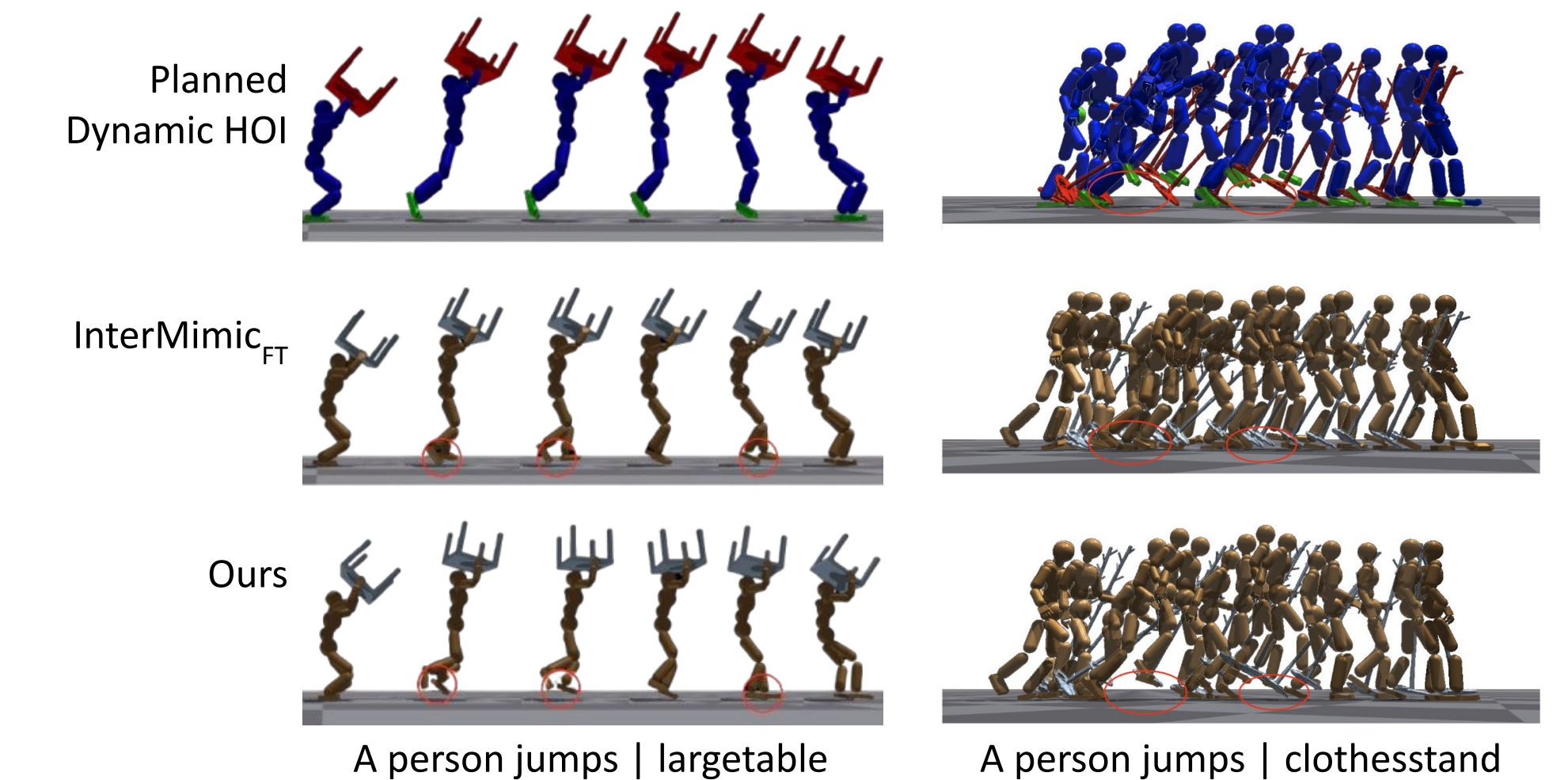
- Diverse object geometries (FullBodyManip)
- Diverse motion styles (push, drag, swing)
- Plug-and-play application — e.g., MDM → GMD enables path planning

Qualitative Results



Comparison of HOI Planning

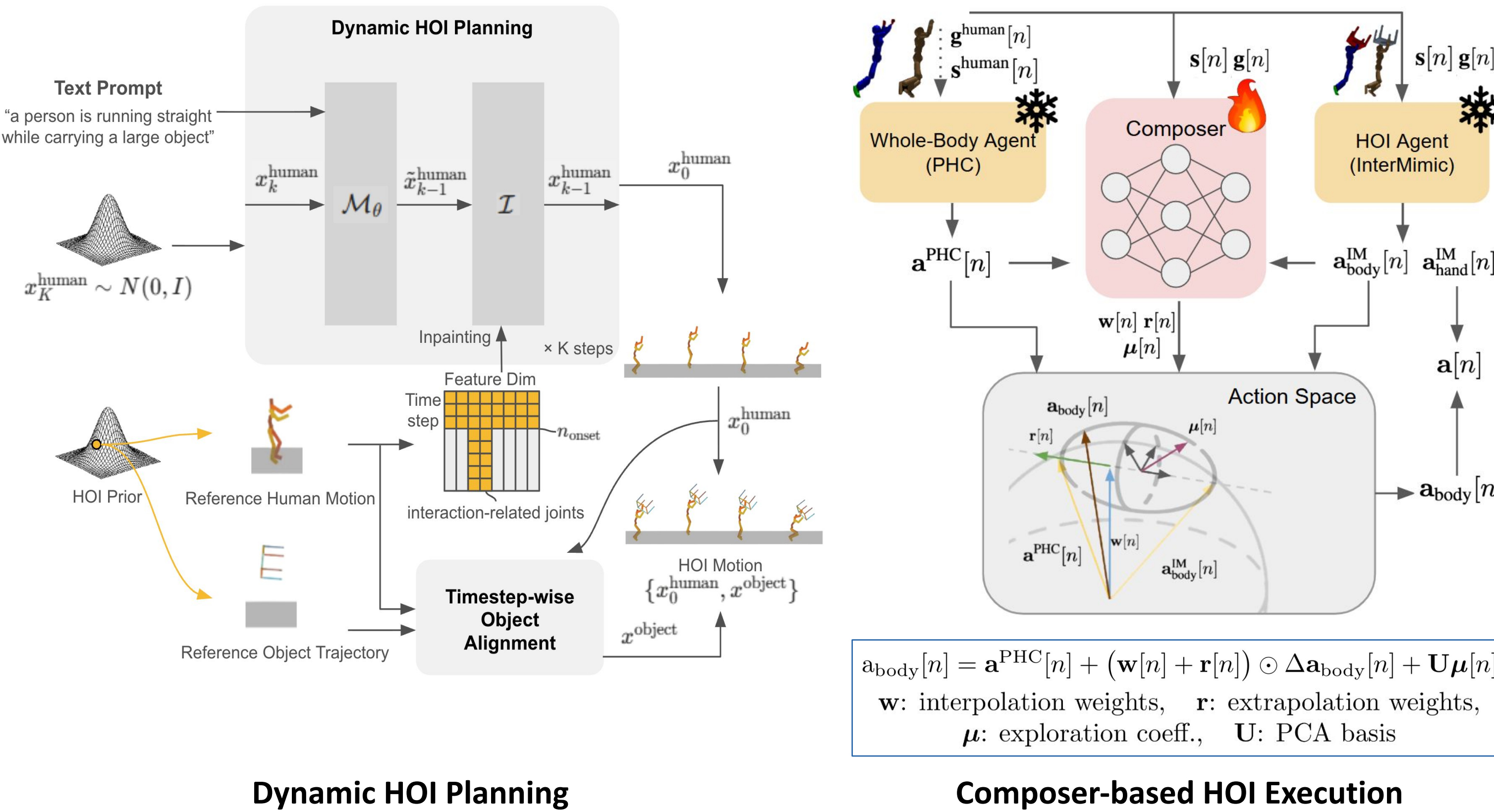
- Ours accurately follows the text-aligned **dynamic motion** while maintaining **consistent hand-object contact**



Comparison of HOI Execution

- Ours successfully executes the planned dynamic HOI while maintaining **motion style** and **object interaction**
- InterMimic_{FT} collapses to a local minimum — shuffling instead of two-foot takeoff

Methodology



Dynamic HOI Planning

- Inpaint HOI prior (FullBodyManip) into MDM during diffusion sampling
- Fix interaction-related joints after contact onset; text-aligned motion inpainting for the others
- Compute object trajectories from hand poses

Composer-based HOI Execution

- Composer predicts per-DoF inter-/extrapolation weights
- PCA-based low-dim subspaces for further exploration

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